Materials Management (MM)

Curriculum: Introduction to S/4HANA using Global Bike



SAP University Alliances

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Teaching material - Information

Teaching material - Version

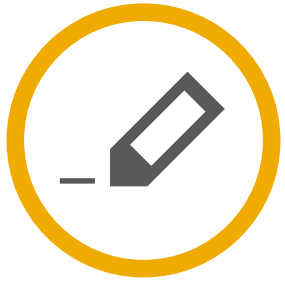
- 4.3 (July 2025)

- Software used
  - S/4HANA 2023
  - Fiori 3.0
- Model
  - Global Bike

- Prerequisites
  - No Prerequisites needed

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Module Information



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Target Audience

- Beginner

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Module Information



Learning Objectives

You are able to:

- Describe the purchase-to-pay business process
- Identify the organizational levels that support the purchase-to-pay business process
- Create a material master record
- Create a vendor master record
- Create a purchase requisition for stock and consumable materials
- Create a purchase order by referencing a purchase requisition
- Post a goods receipt for a purchase order
- Explain the warehouse structure in Warehouse Management (WM)
- Enter a vendor's invoice
- Process the automatic payment run
- Explain the integration points of the purchase-to-pay business process

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Functionality

- Inventory Management
- Purchasing
- MRP
- Physical Inventory
- Valuation
- Service Master
- Invoice Verification
- Product Catalogs

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Unit Overview

- MM Organizational Structure
- MM Master Data
- MM Processes
  - Procure-to-Pay Process

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MM Organizational Structure (Materials Mgmt.)

- Client
  - An independent environment in the system
- Company Code
  - Smallest organizational unit in business terms for which complete, self-contained accounting (balance sheet, income statement, etc.) can be maintained
- Plant
  - Operating area or branch within a company
    - Manufacturing, distribution, purchasing or maintenance facility
- Storage Location
  - An organizational unit allowing differentiation between the various stocks of a material in a plant

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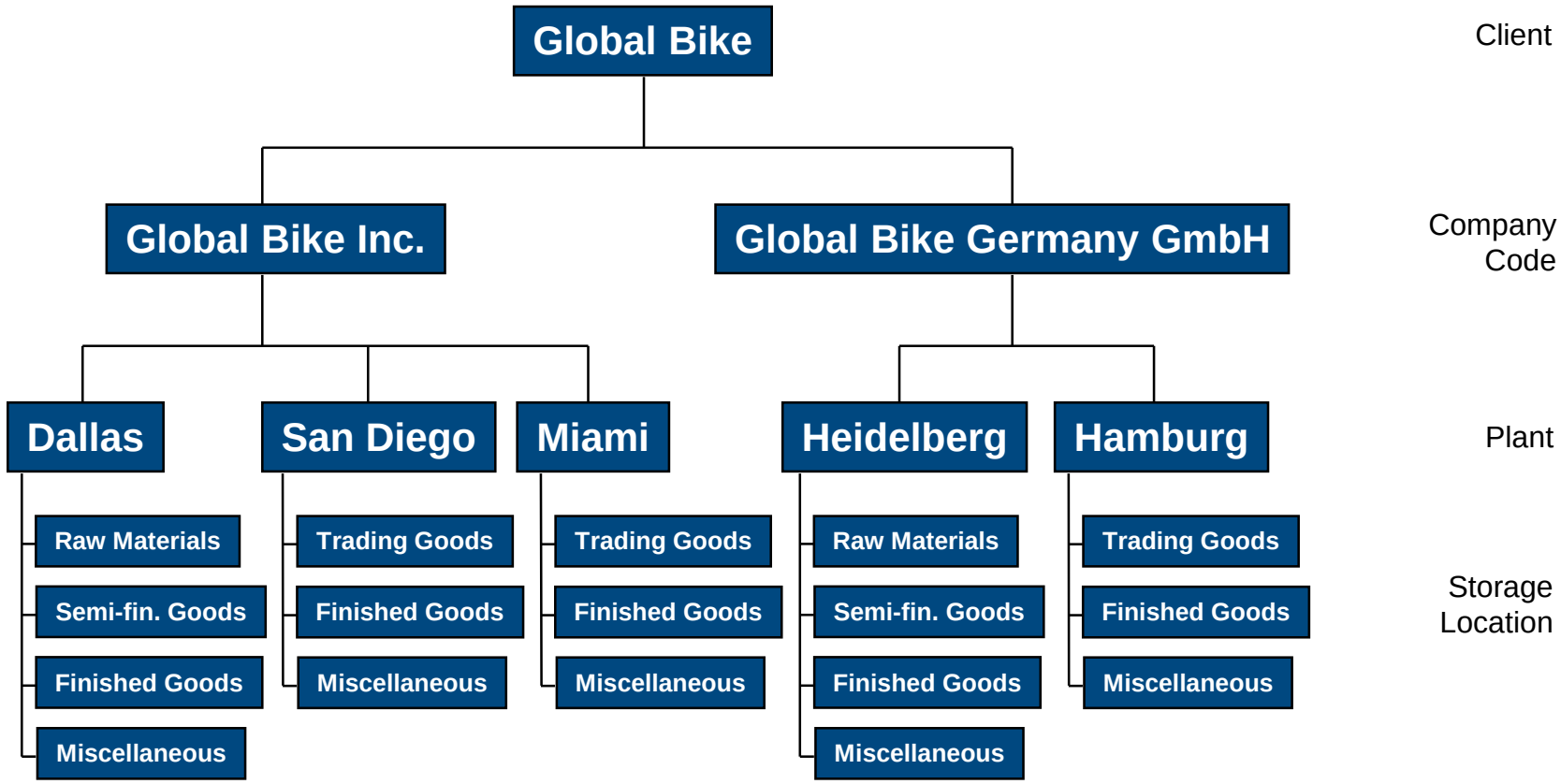
MM Organizational Structure (Purchasing)

- Purchasing Organization
  - The buying activity for a plant takes place at the purchasing organization
  - Organization unit responsible for procuring services and materials
  - Negotiates conditions of the purchase with the vendors

- Purchasing Group
  - Key that represents the buyer or group of buyers who are responsible for certain purchasing activities
  - Channel of communication for vendors

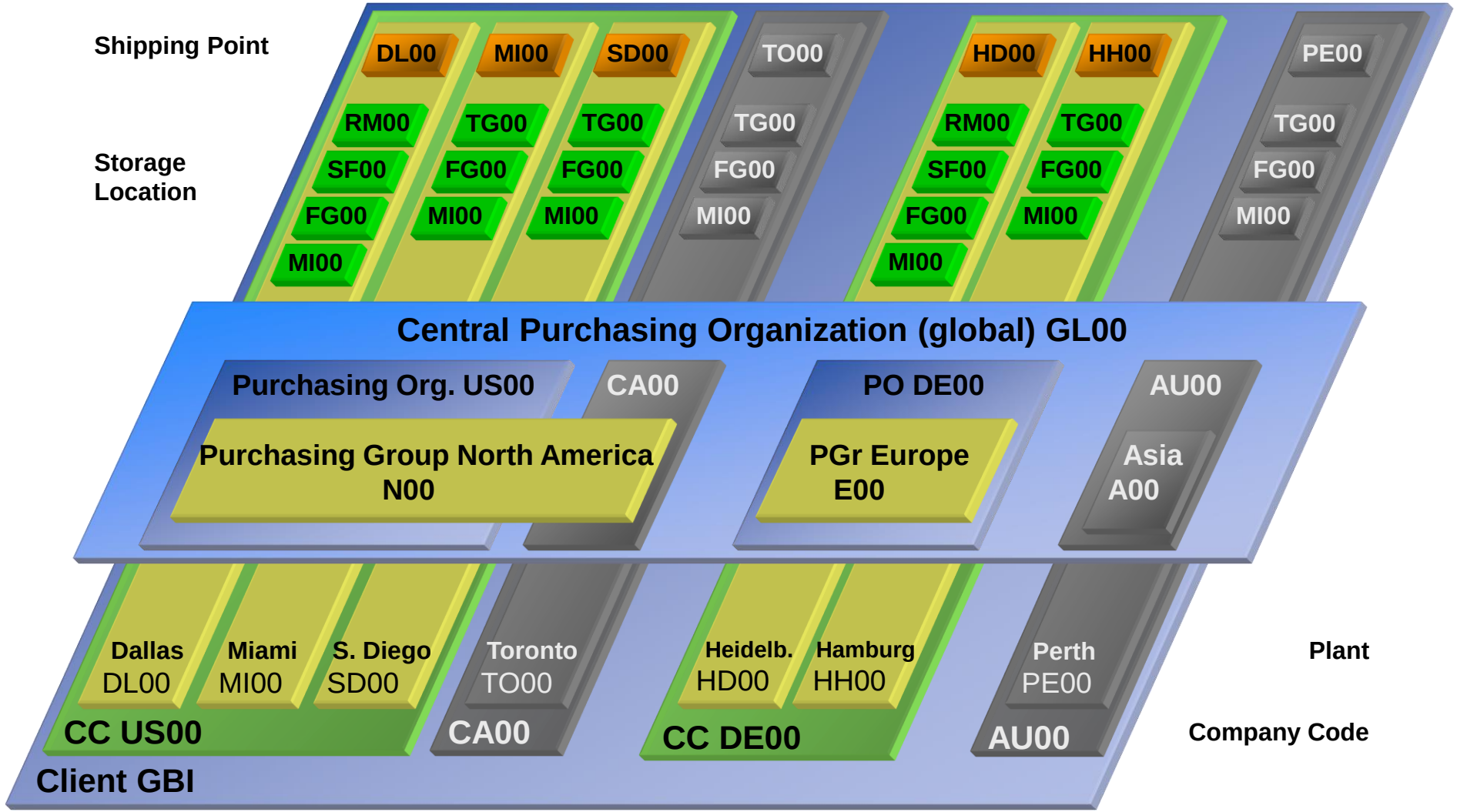
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Global Bike Structure for Materials Management



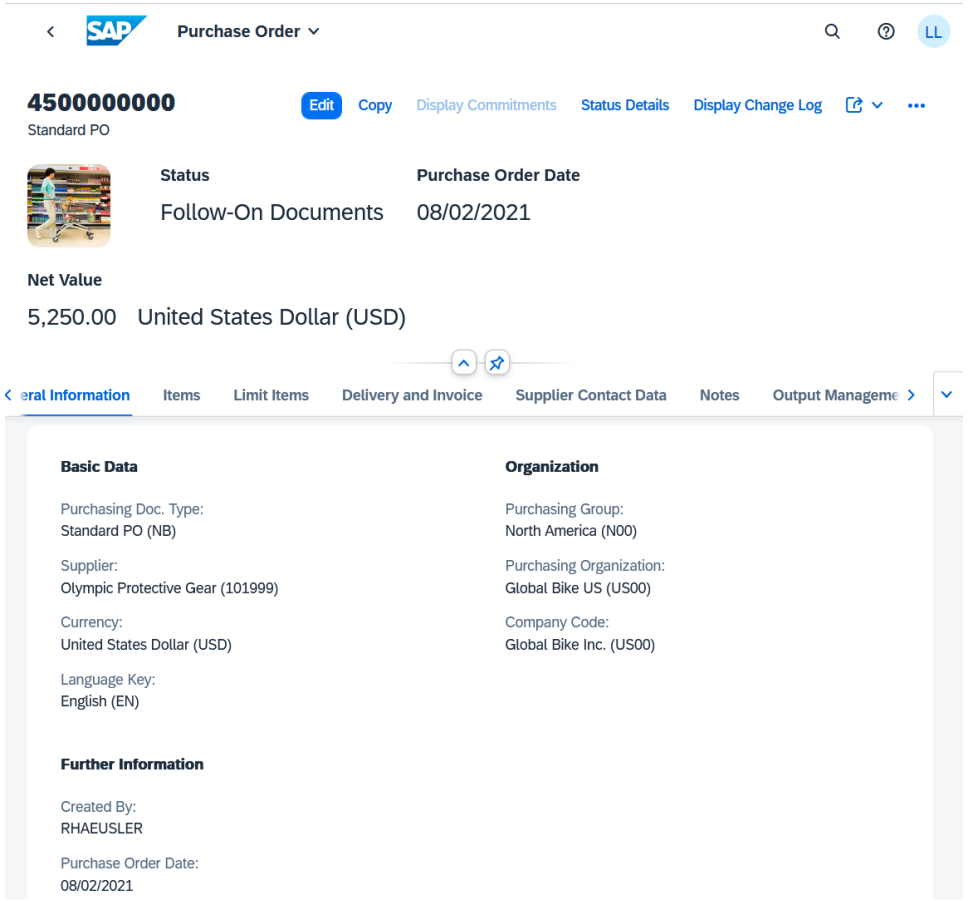
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Global Bike Enterprise Structure in SAP ERP (Logistics)



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MM Master Data

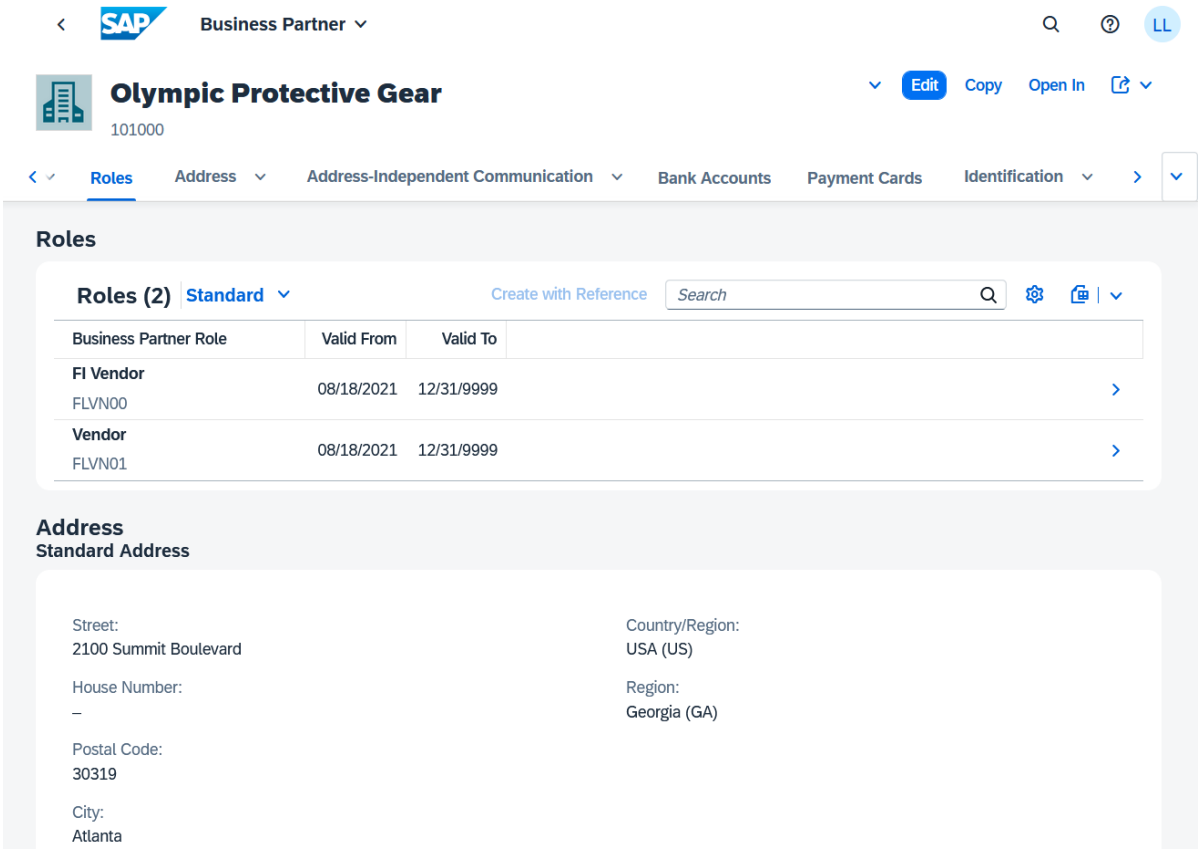


- Vendor Master Data
- Material Master Data
- Purchasing Info Record
- Condition Master Data
- Output Master Data

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Vendor Master Data

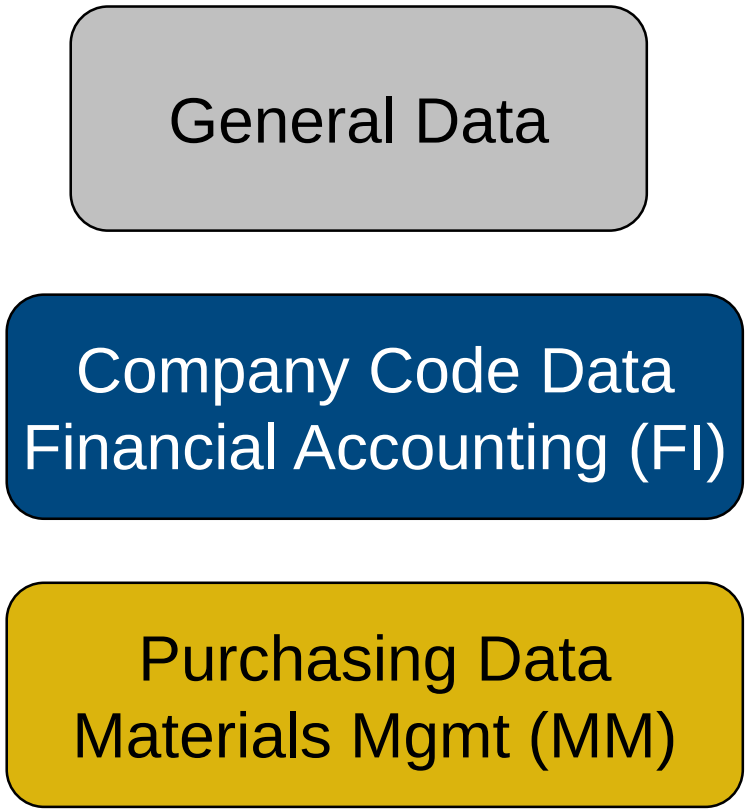
- Vendor Master
  - Contains all the necessary information needed to business with an external supplier
  - Used and maintained primarily by the Purchasing and Accounting Departments
  - Every vendor MUST have a master record
  - The supplier master record is created in SAP S/4HANA as a central business partner master record with different roles.
  - For suppliers, these are often the business partner roles:
    - FLVN00 (FI Vendor)
    - FLVN01 (Vendor)



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Vendor Master Views

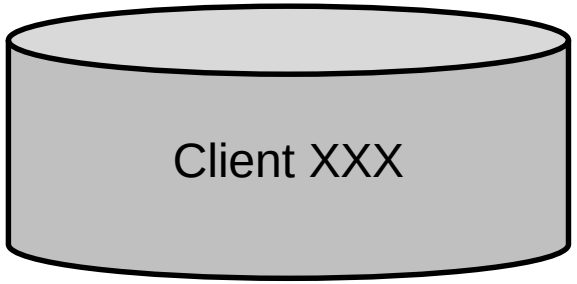
- Client Level
  - Address
  - Vendor Number
  - Preferred Communication
- Company Code Data
  - Reconciliation Account
  - Terms of Payment
  - Bank Account
- Purchase Org Data
  - Purchasing Currency
  - Salesman’s Name
  - Vendor Partners



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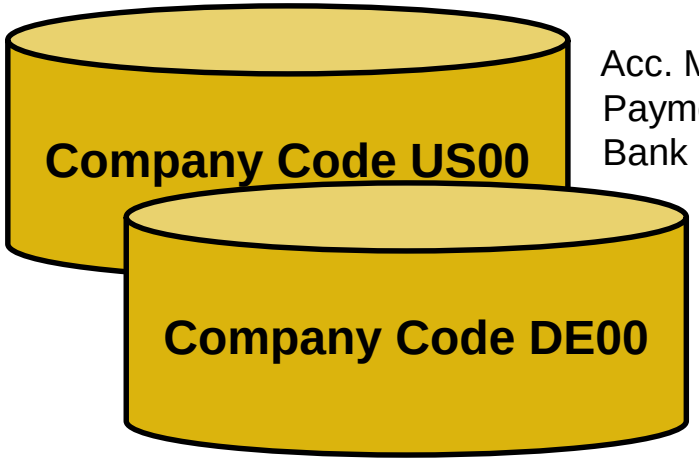
Vendor Master

General Information relevant for the entire organization:



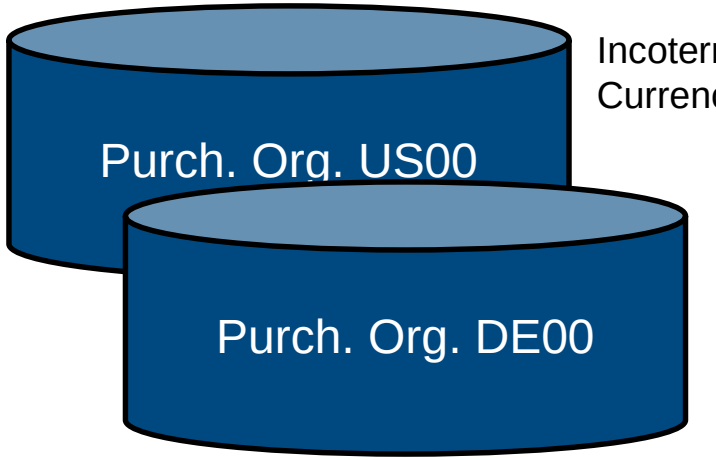
Name
Address
Communication

Company Code specific information:



Acc. Mgmt
Payment
Bank

Purch. Organization specific information:

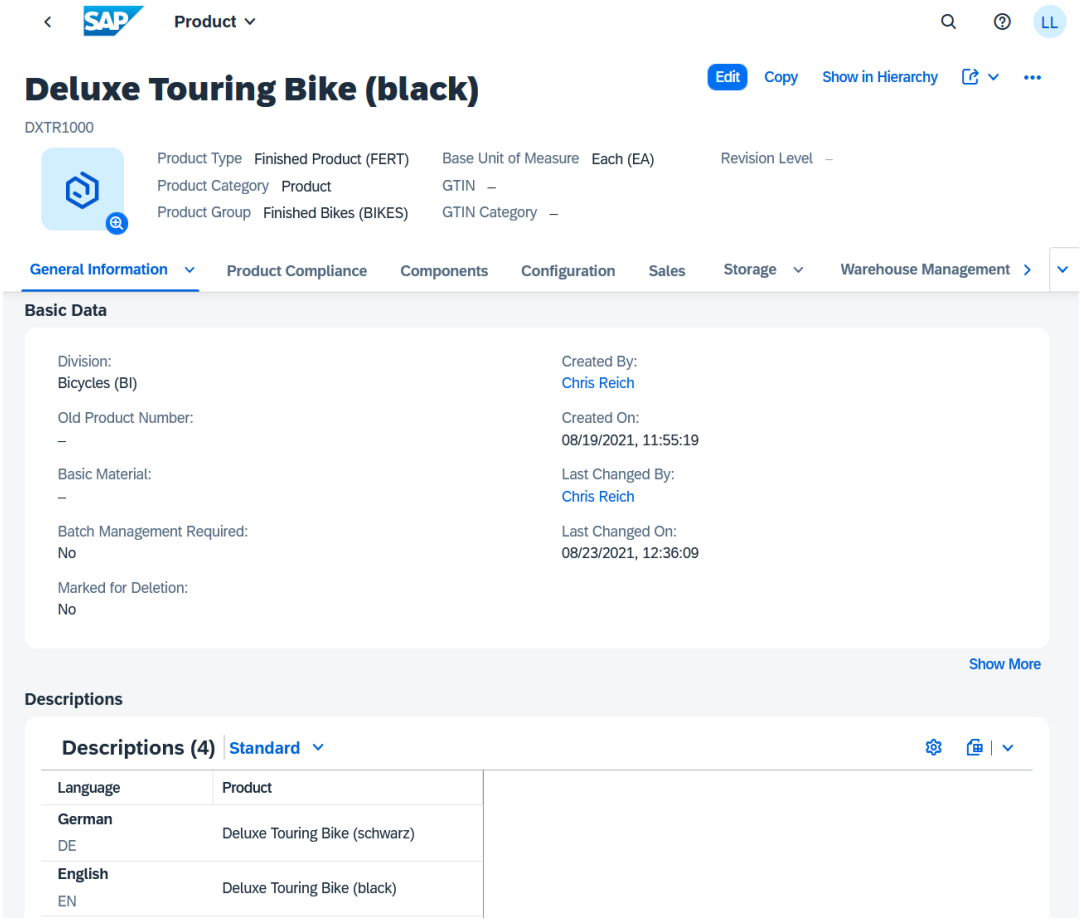


Incoterms
Currency

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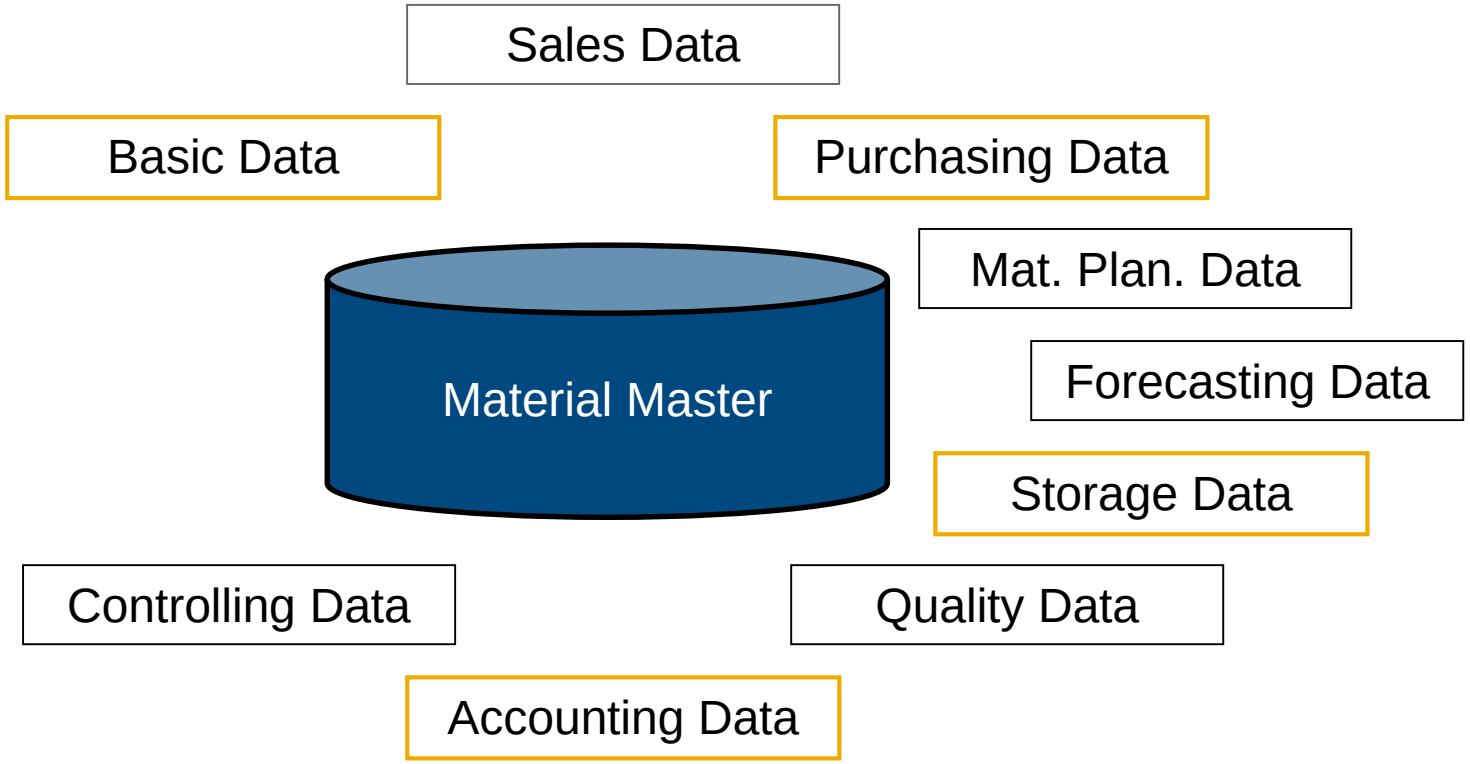
Material Master Data

- Material Master
  - Contains all the information a company needs to manage about a material
  - It is used by most components within the SAP system
    - Sales and Distribution
    - Materials Management
    - Production
    - Plant Maintenance
    - Accounting/Controlling
    - Quality Management
  - Material master data is stored in functional segments called Views



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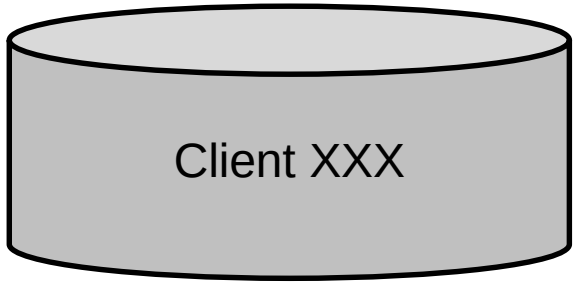
Material Master Views



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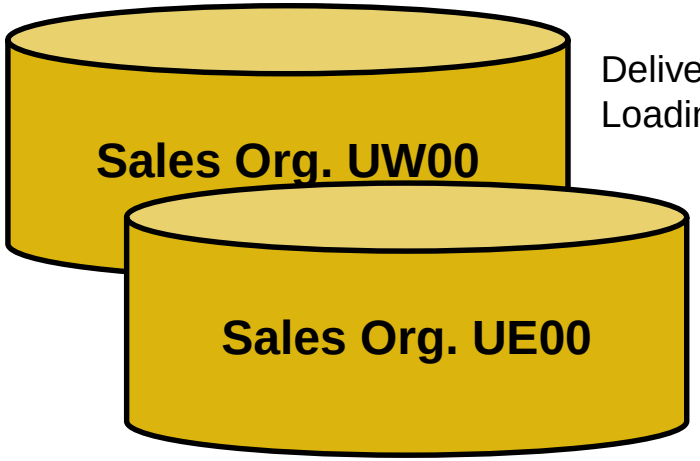
Material Master

General Information relevant for the entire organization:



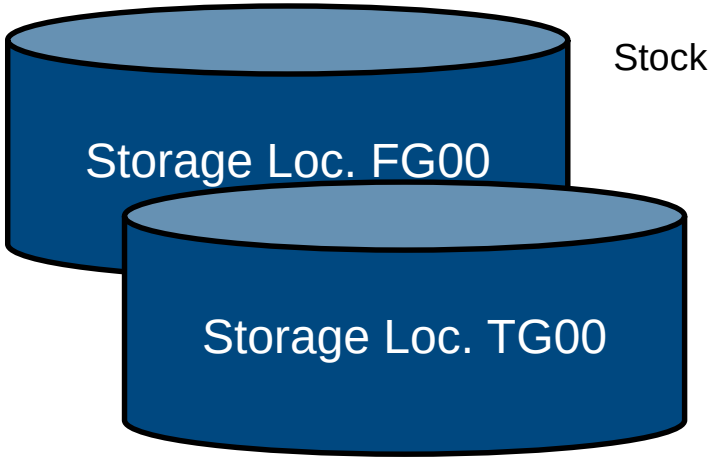
Name
Weight
Unit of Measure

Sales specific information:



Delivering Plant
Loading Grp

Storage Location specific information:

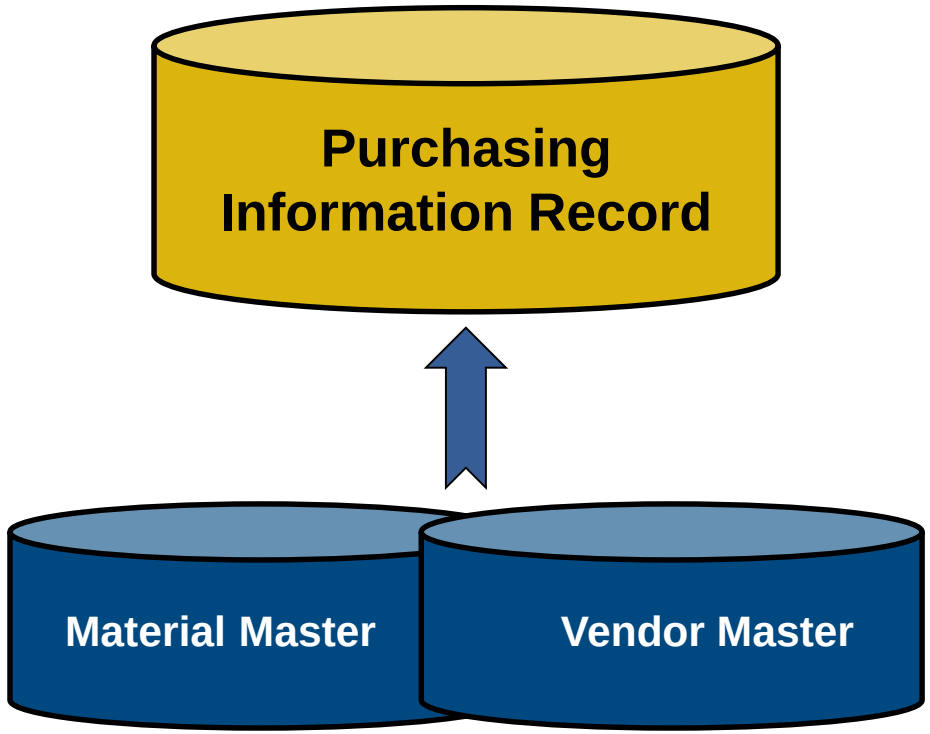


Stock Qty

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Purchasing Information Record

- Framework for Purchase Order
  - Contains the relationship between a vendor and a material
- Can be created:
  - Manually
  - Automatically – Quotations
  - Automatically – Purch. Orders
- Reporting
  - Vendor Evaluation



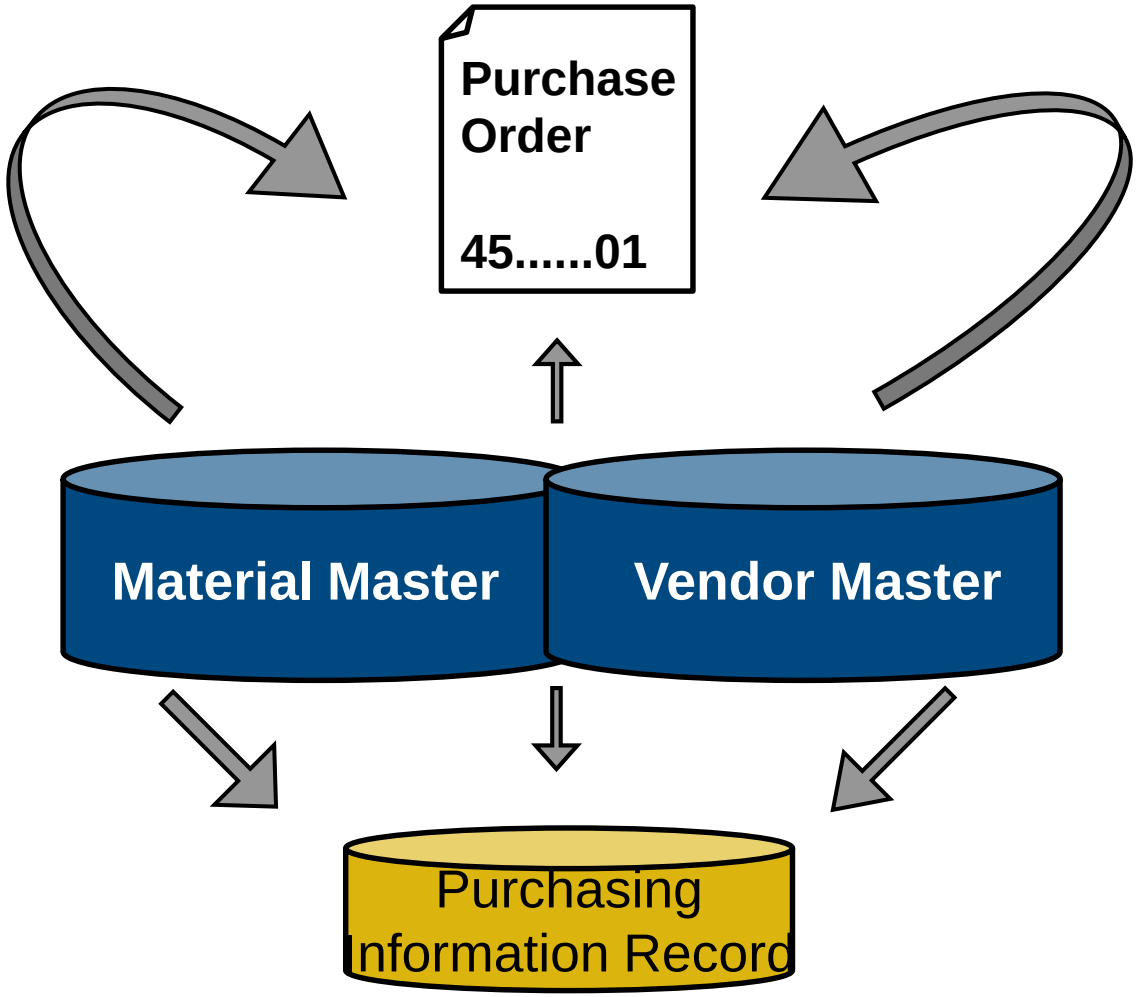
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Purchasing Information Record

- Allows buyers to quickly determine:
  - Which vendors have offered or supplied specific materials
- Info Records contain:
  - Data on pricing and conditions
  - Last purchase order
  - Tolerance limits for deliveries
  - Specific lead times
  - Availability periods
  - Vendor Evaluation data
- Serves as default information for Purchase Orders

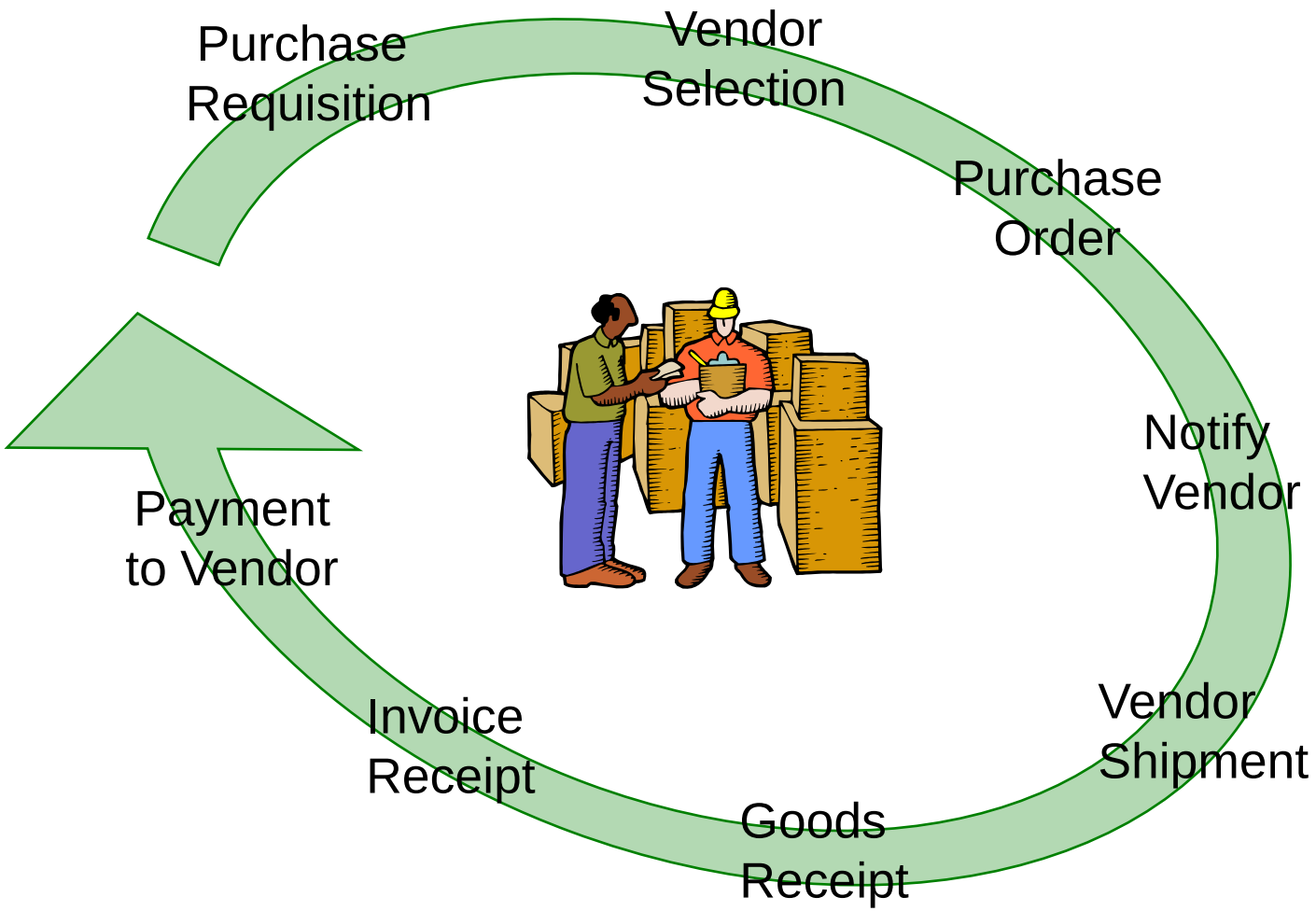
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Master Data in Use



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Procure-To-Pay Process



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Purchase Requisition

- Internal Document instructing the purchasing department to request a specific good or service for a specified time

- Requisitions can be created two ways:
  - Directly - Manually
    - person creating determines: what, how much, and when
  - Indirectly - Automatically
    - MRP
    - Production Orders
    - Maintenance Orders
    - Sales Orders

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Requisition Sourcing

- Once the requisition has been assigned a source of supply it can be released for processing
- There are a variety of ways that a purchasing department can process a requisition to determine the appropriate Source of Supply:
  - Internal Sourcing Requirements
  - Source List
  - Outlined Agreement
  - RFQ

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Internal Sourcing

- The requisition for materials could be satisfied by sources within our company.
  - It is possible that a plant within your firm could represent a potential source of supply for the material needed (centralized warehouse)
  - If an internal source is identified the requirement is covered by an internal procurement transaction (stock transport order)

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Source List

- A source list is a record that specifies the allowed means for procuring a material for a certain plant within a given time period.
  - If the list contains a sole source the system will assign the vendor to the requisition.
  - If several options exist the system will display a list of vendors for you to choose from.
  - If no source has been established the system will revert to search information records and outline agreements.

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Outline Agreement

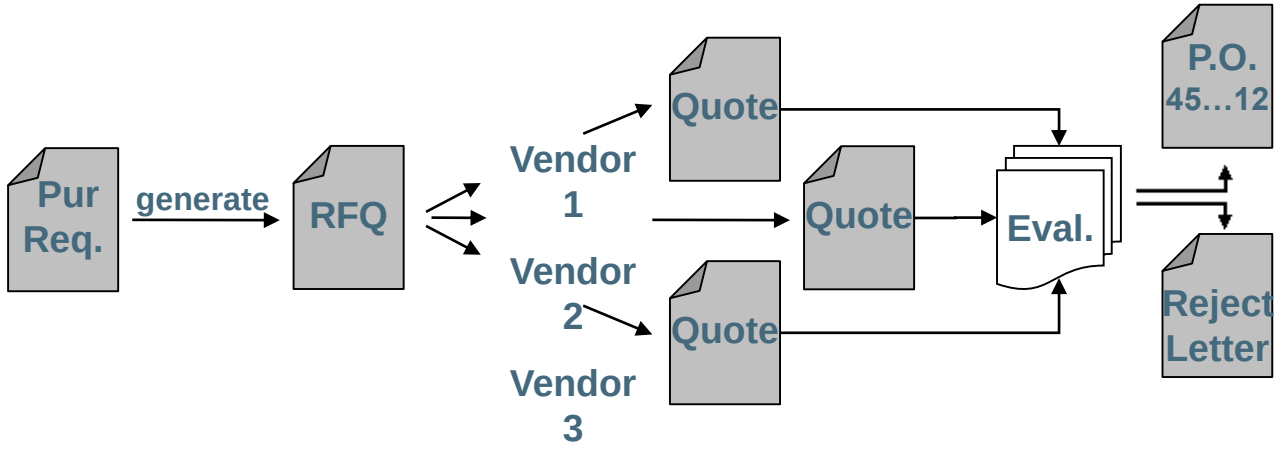
- Requisitions can be satisfied through existing longer-term purchasing agreement

- These agreements are subdivided into:
  - Contracts
    - Consists of items defining the individual materials, material groups, or services with prices and in many cases quantities
      - Quantity
      - Value
  - Scheduling Agreements
    - Total quantity of material is spread over a certain period in a delivery schedule, consisting of line items indicating quantities and their planned delivery date

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Request for Quotation

- If nothing exist in the system we may need to submit a request for quotation to our vendors. An RFQ is an invitation to a vendor by a Purchasing Organization to submit a bid for the supply of materials or services
  - The accepted quotations will generate Purchasing Information Records
  - Perform Quotation Price Comparisons
  - Finally Select a Quotation



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Quotation from Vendor

- The quotation received by your company is a legally binding offer, should decide to do business with the vendor, containing price’s and conditions for the materials specified in the RFQ for a predefined period of time.
  - In SAP the RFQ and the Quotation will be become a single document, you will enter the vendor’s response in the RFQ you created.

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Vendor Evaluation once Identified

- Vendor evaluation helps purchasing evaluate vendors for sourcing while also enabling the company to monitor vendor relationships through performance scores and criteria you put in place.
  - Supports a maximum of 99 main criteria and 20 sub criteria for each main:
    - Price
      - Price Level
      - Price History
    - Quality
      - Goods Receipt
      - Quality Audit
      - Complaints/Rejection level
    - Delivery
      - On-time delivery performance
      - Quantity reliability
      - Compliance with shipping instructions
      - Confirmation Date
  - You then must establish a scoring range (1 -100) and determine the weight factors of scores for each.

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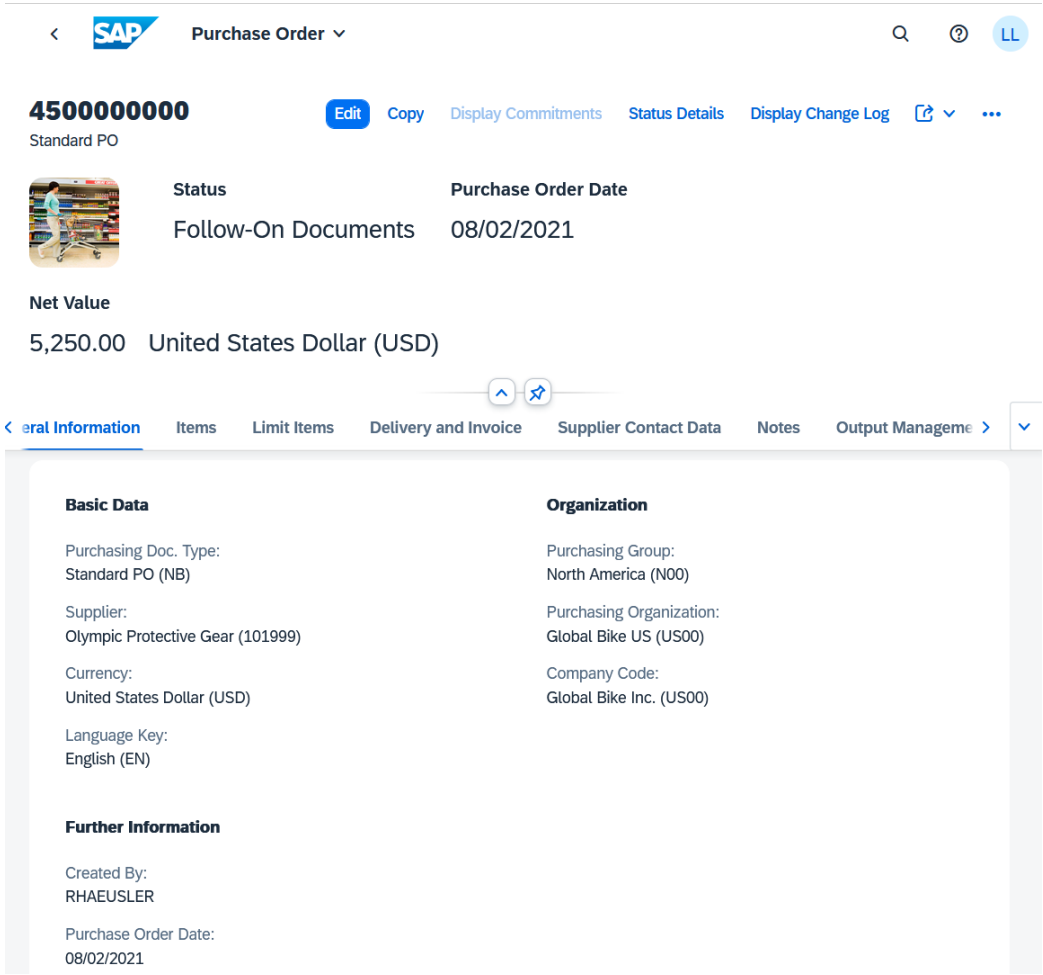
Purchase Order

- A purchase order is a formal request to a vendor for a specific material or service under the stated conditions
- Purchase Orders can be created manually
  - Reference a Purchase Order
  - Reference a Purchase Requisition
  - Reference a RFQ/Quotation
  - Without Reference
- Purchase Orders can be create automatically

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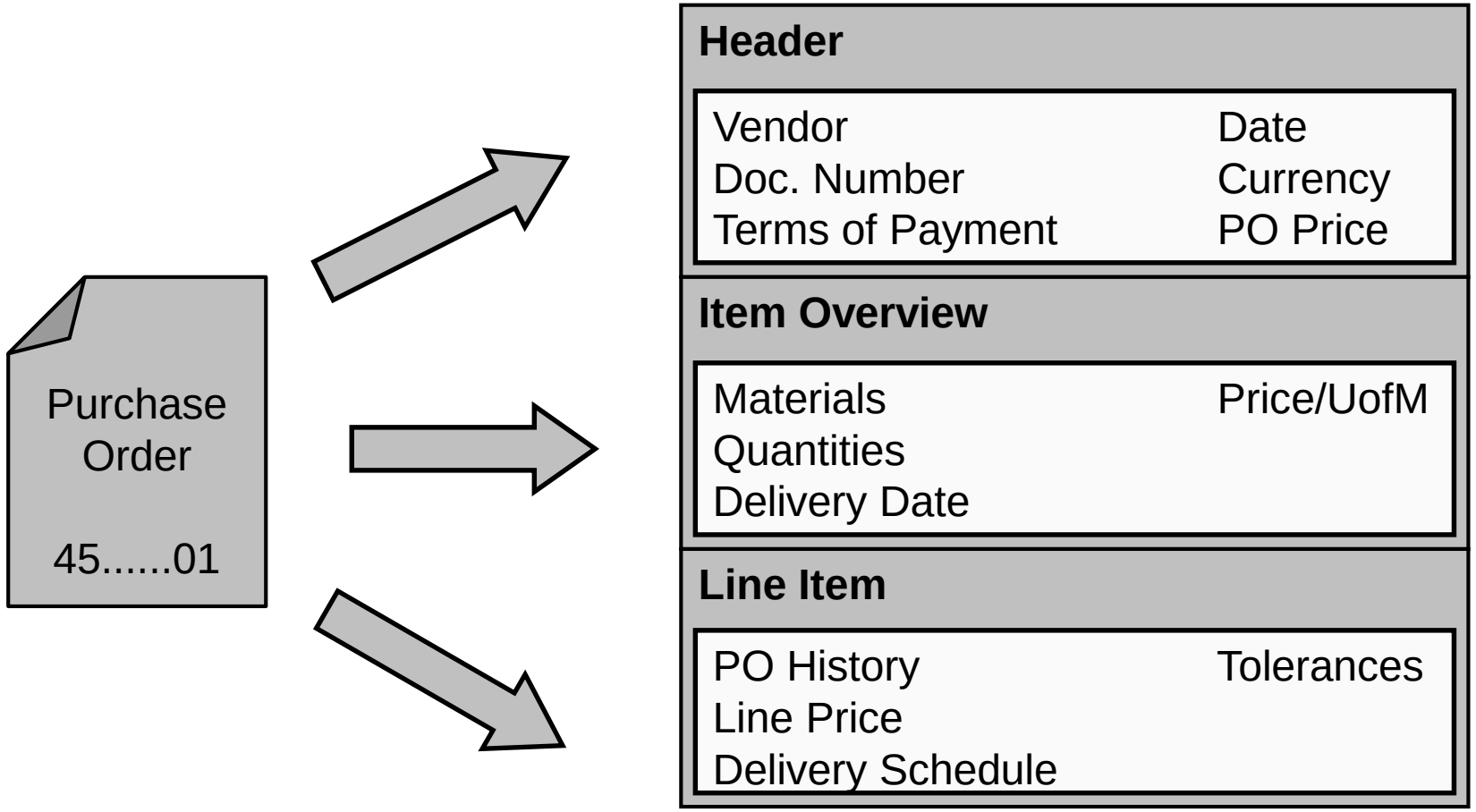
Purchase Order

- A purchase order can be used for a variety of purposes, the item category (procurement type) defined in the PO will dictate the use of the order and the process that the order will follow:
  - Standard
    - Stock or Consumption
  - Services
  - Subcontracting
  - Third-Party
  - Consignment



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Purchase Order Structure



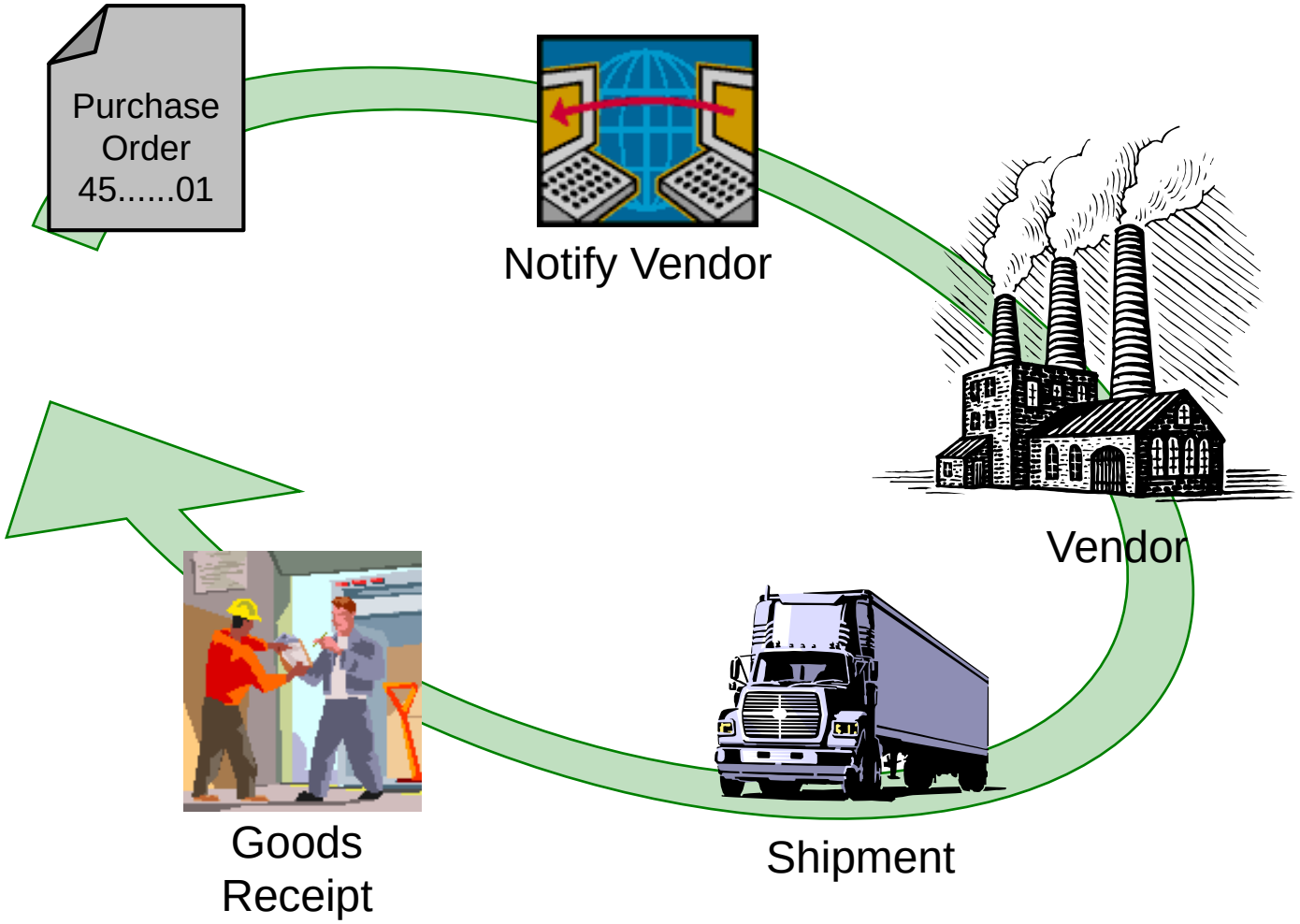
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Purchase Order Output

- Once a Purchase Order has been created the vendor needs to be notified
  - Printed
  - E-mail
  - EDI
  - Fax
  - XML
- There are a variety of forms that aid in the purchasing process and are generated from the Purchase Order
  - Purchase Order Output
  - Order Acknowledgement Forms
  - Reminders
  - Schedule Agreements

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Goods Receipt



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Goods Receipt

- Goods movement in which we accept goods into our system

- If materials are delivered against a Purchase Order we will reference that Order:
  - Determine if we got what we ordered
  - System can purpose data for us from the PO
    - Material, quantity
  - Purchase Order History is update with the receipt
  - Updates Physical Inventory
  - Updates Inventory G/L Account

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Material Movements

- When a goods movement takes place it is represented by a Movement Type

- Movement types are three-digit keys used to represent a movement of goods
  - 101 – goods receipt into warehouse
  - 103 – goods receipt into GR blocked stock
  - 122 – return delivery to vendor
  - 231 – consumption for a sales order
  - 561 – initial entry of stock

- Destinations for Receipt of Goods
  - Warehouse – Unrestricted, Quality, Blocked
  - Quality
  - Goods Receipt Blocked Stock

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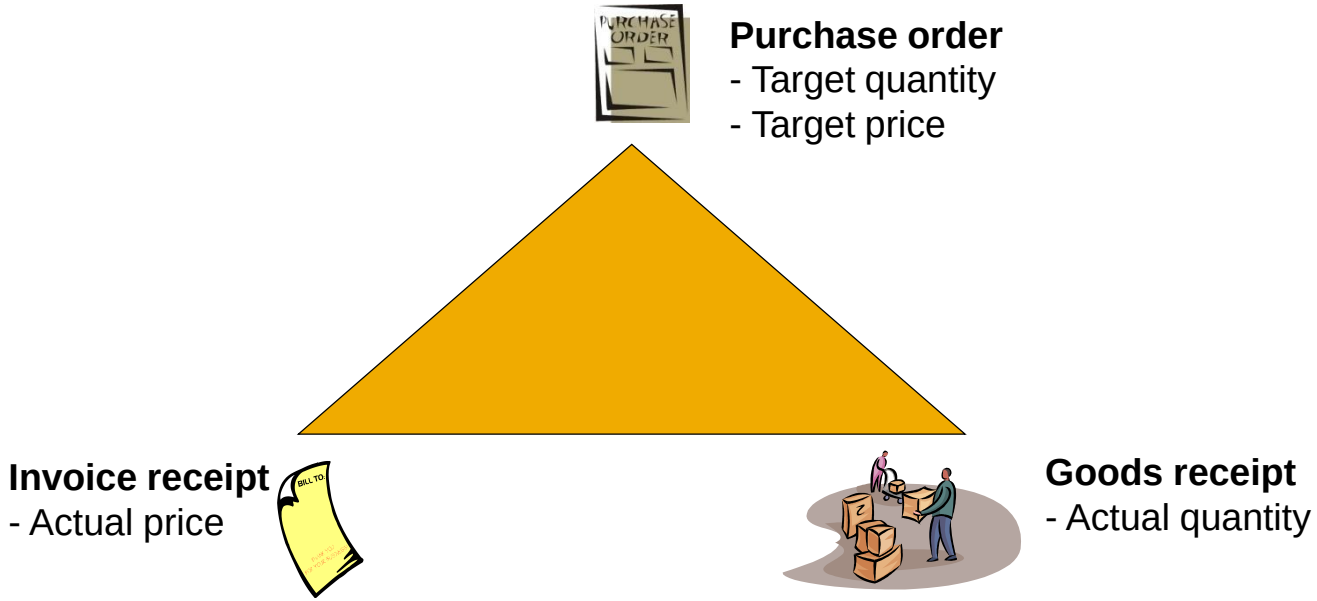
Effects of a Goods Receipt

- When a Goods Movement for the receipt of goods takes place a series of events occur
  - Material Document is Created
  - Accounting Document is Created
  - Stock Quantities are Updated
  - Stock Values are Updated
  - Purchase Order is Updated
  - Output can be generated (GR slip / pallet label)

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Invoice Processing

- Incoming Invoices are reference against a Purchase Order to verify their content, prices, and arithmetic.
- If discrepancies arise between the purchase order or goods receipt and the invoice the system with generate a warning or an error
  - Depending on system configuration the difference could cause the system to Block the Invoice



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Invoice Processing

- When an invoice is saved it applies the liability from the Goods Receipt of our Purchase Order to a Vendor
- Upon verification the:
  - Purchase Order is updated
  - Material Master is Updated (MAP)
  - Accounting Document is created
- Once the Invoice has been posted the verification process is completed and the payment process is initiated within Financial Accounting

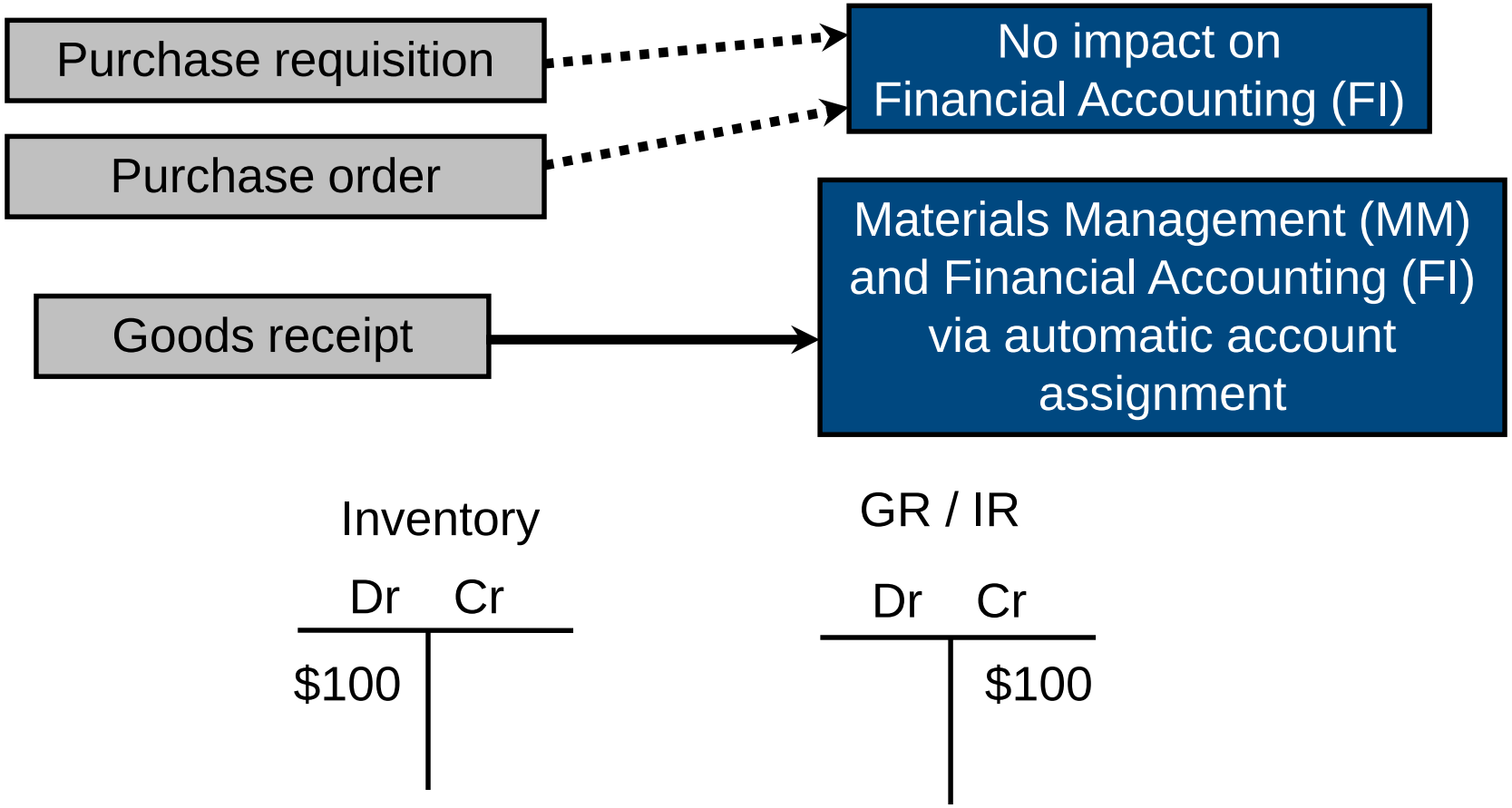
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Payment to Vendor

- Can be done automatically or manually
  - Post Outgoing Payment vs. Payment Program
- Elements of the Payment Transaction:
  - Payment Method
  - Bank from which they get paid
  - Items to be Paid
  - Calculate Payment Amount
  - Print Payment Medium
- Process will create a financial accounting document to record the transaction

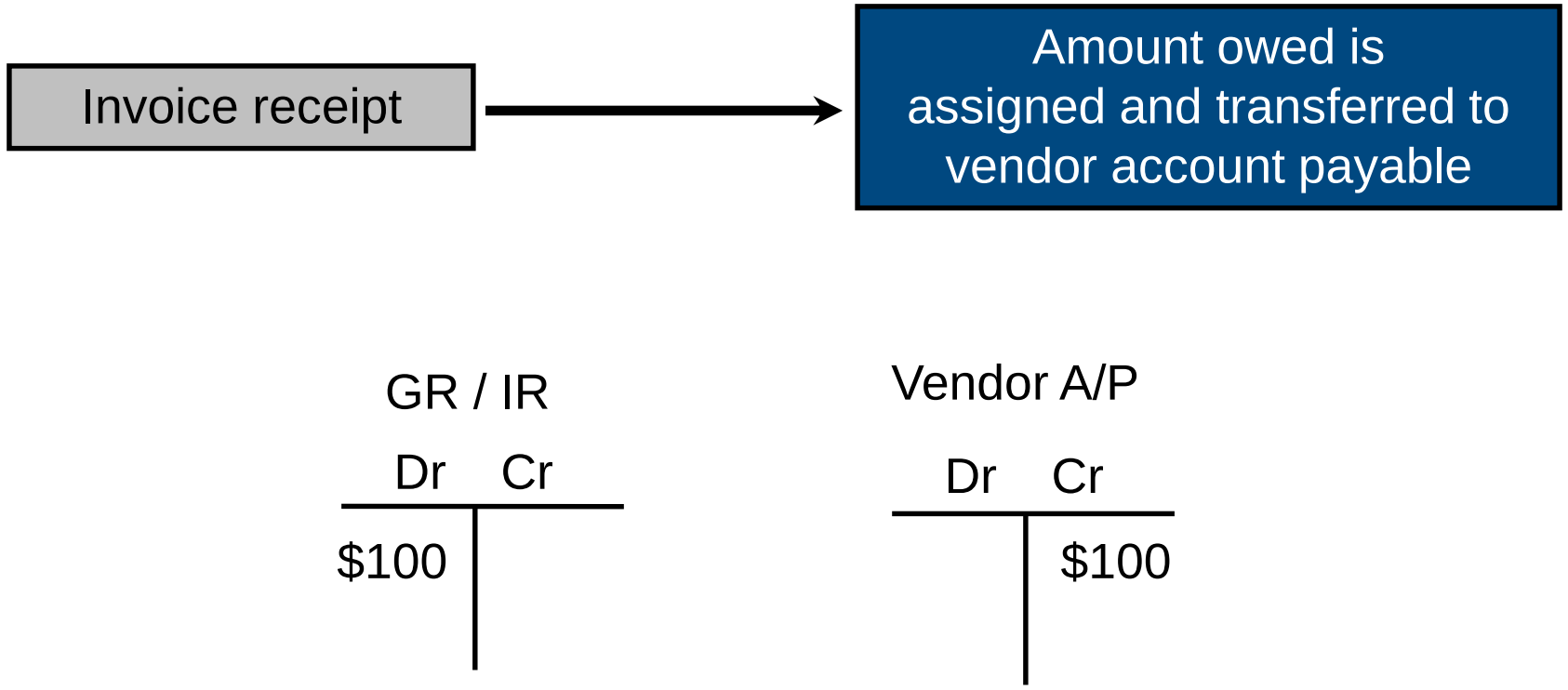
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Goods Receipt / Invoice Receipt Reconciliation Account



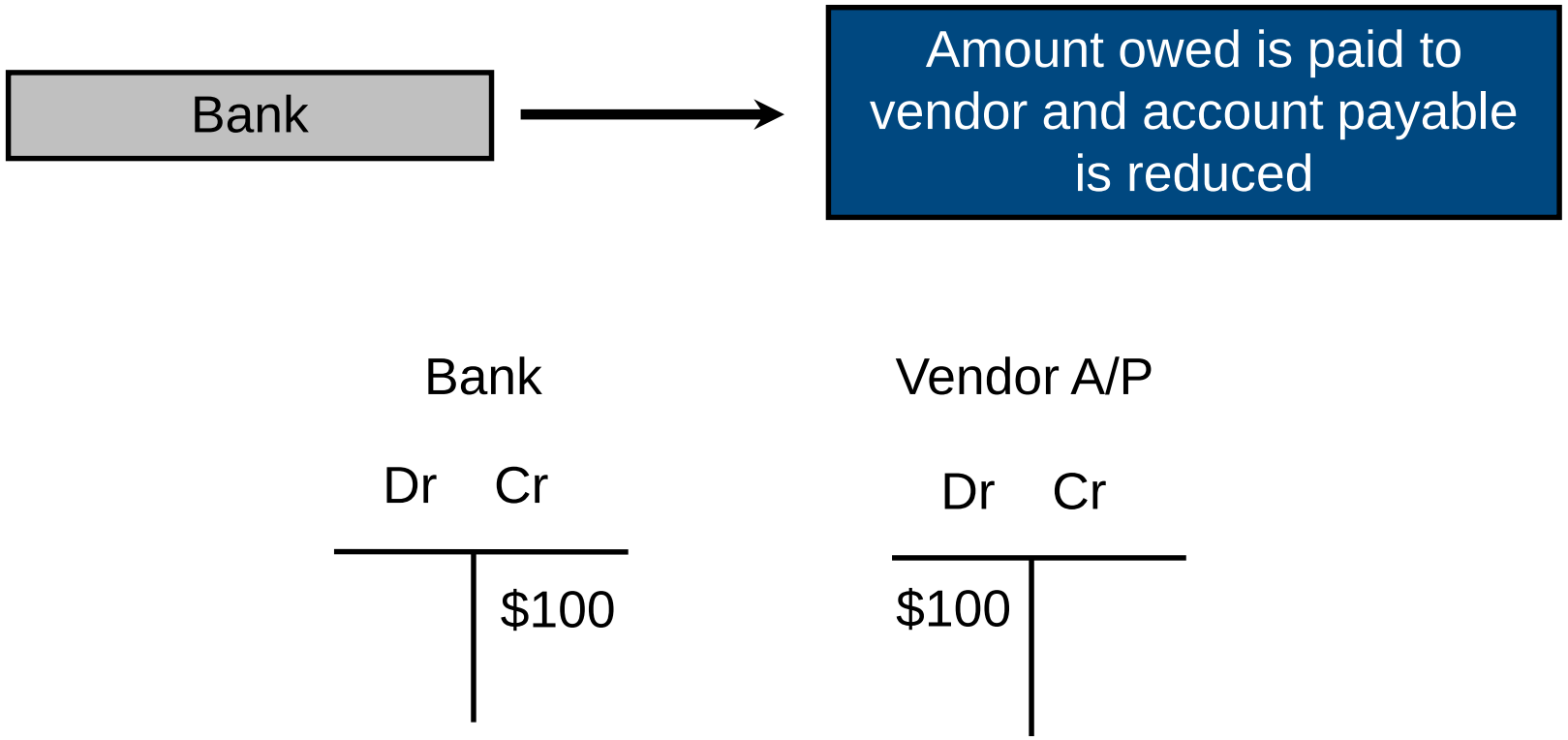
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Goods Receipt / *Invoice Receipt* Reconciliation Account



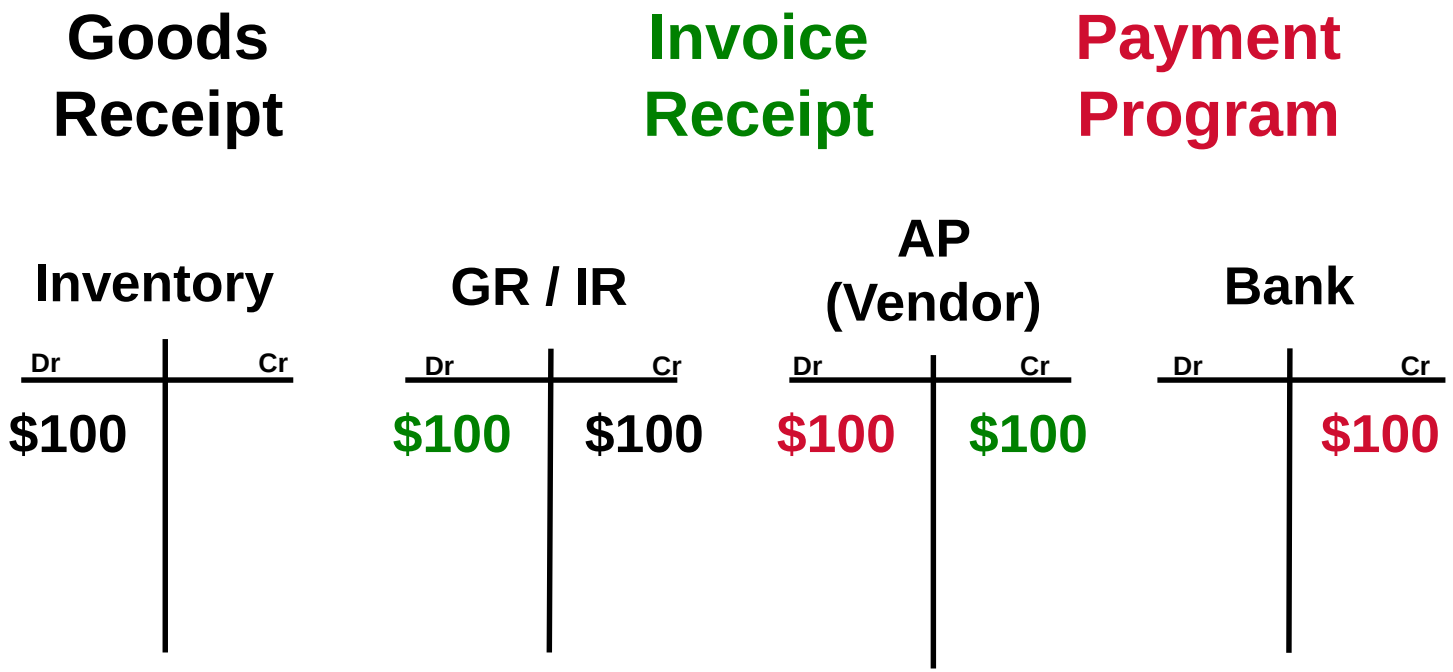
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Vendor Payment



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FI – MM Integration Point



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